Design of an Integrated Circuit (IC) for Comparison of Two Binary Digits

You are required to design a transistor-level Integrated Circuit (IC) that compares **two single-bit binary inputs** and determines their logical relationship.

- **A > B**
- **A = B**
- **A < B**

The design must integrate multiple logic families, satisfy low-power and high-speed constraints, and be fully validated through simulation.

---

1. Objective

Design a digital IC capable of accepting two binary inputs **A** and **B**, each of value 0 or 1, and determining the magnitude relationship between them. The output must be glitch-free, power-efficient, and optimized for minimum area.

---

2. Input Interface Requirements

Your IC will take the following inputs:

- **A**: 1-bit binary input
- **B**: 1-bit binary input

Inputs may be given using **switches, VPulse signals, or digital sources**.

---

3. Logical Comparison Block

Implement transistor-level logic to evaluate:

- **A > B:** occurs when A=1 and B=0
- **A = B:** occurs when A=B (00 or 11)
- **A < B:** occurs when A=0 and B=1

Each expression must be derived and implemented using a combination of logic families.

---

4. Logic Family Requirements

You must use **two of the following logic families**:

- CMOS
- Complex CMOS
- Dynamic Logic

You must justify all design decisions.

---

5. Simulation Requirements

Using Cadence or LTspice, provide:

- Complete schematic diagram
- Transient simulation results
- Propagation delay measurements
- Verification of all four possible input combinations:
    - A=0, B=0
    - A=0, B=1
    - A=1, B=0
    - A=1, B=1

Waveforms must clearly show the correct one output behavior.

---

6. Documentation Requirements

Your report must include:

- Front page
- Introduction
- Project requirements (detailed)
- Working principle of binary comparator IC
- Logic expressions and truth table
- Schematic diagrams
- Simulation results

- Conclusion with individual contributions

---

7. Expected Output Behavior

A	B	A>B	A=B	A<B
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

Corresponding one-hot outputs must be stable and correct under all combinations.